Math 141, S06 Name: key

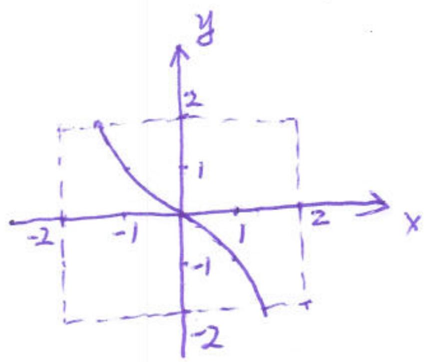
Quiz 1 Section: SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1.(5 points) Express the function $f(x) = -x|x|$ in piecewise form without using absolute values and draw its graph in the viewing window $[-2, 2] \times [-2, 2]$.

Solution:

$$f(x) = -x|x| = \begin{cases} -x^2, & x \geq 0 \\ x^2, & x < 0 \end{cases}$$



Problem 2.(5 points) Find the natural domain and range of the function $f(x) = \sqrt{2 - x^2}$.

Solution: It requires

$$2 - x^2 \geq 0 \quad - ①$$

$$x^2 \leq 2$$

$$-\sqrt{2} \leq x \leq \sqrt{2}$$

So the domain of $f(x)$ is $[-\sqrt{2}, \sqrt{2}]$.

On the other hand, we know $x^2 \geq 0$, so

$$2 - x^2 \leq 2. \quad - ②$$

Combining ① & ②, we get

$$0 \leq \sqrt{2 - x^2} \leq \sqrt{2}$$

So the range of $f(x)$ is $[0, \sqrt{2}]$.